

Build a Shopify Discount App

From zero to a live public app on the Shopify App Store.
A step-by-step learning guide for complete beginners.

20
Steps

4
Phases

8+
Discount Types

1
Public App

DiscountLab App · Learning Guide · Version 1.0

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Understanding Shopify Apps

Before writing a single line of code, you need to understand what a Shopify app actually is, how it connects to Shopify, and the different types that exist.

The Mall Analogy

Think of Shopify like a shopping mall. Each merchant's store is a shop inside that mall. Your app is like a service kiosk that merchants can hire to work inside their shop. Just like a real mall has different types of kiosks — permanent shops, pop-ups, and vendor carts — Shopify has different types of apps.

The 3 Types of Shopify Apps

Public App

(What you are building)

YOU ARE BUILDING THIS

Listed on the Shopify App Store. Any merchant in the world can find and install it. Revenue comes from subscriptions and freemium plans. Shopify reviews it before listing. This is the highest effort but also the highest reward.

Custom App

Built for one specific store only. Not listed anywhere. A developer builds it exclusively for one merchant's needs. Revenue is a one-time project fee. No App Store review required.

Sales Channel App

A special type of public app that adds a new place to sell — like Instagram, TikTok, or Google Shopping. More complex to build. Subject to a stricter review process.

What 'Embedded' Means

When a merchant installs your app, they don't go to a separate website. Your app loads right inside the Shopify admin panel. The merchant never leaves Shopify. This works using an iframe — Shopify cuts out a portion of its admin page and loads your website inside that space. Your app runs on your server (Railway), but the merchant sees it as if it's part of Shopify. This is what 'embedded app' means.

How Installation Works — The OAuth Flow

When a merchant clicks Install on your app in the App Store, here is exactly what happens:

#	What happens	Details
1	Merchant clicks 'Install'	On the Shopify App Store listing page
2	Shopify redirects	To your app's server URL on Railway
3	Your app asks for permissions	Read products, create discounts, etc.
4	Merchant approves	Clicks Allow in the Shopify UI
5	You receive an access token	A secret key — save it to your database
6	App is installed and ready	Merchant can now use your app

Key insight: The access token is like a master key. Every time your app wants to read or write data from a merchant's store, it shows Shopify this token. The Shopify CLI + Remix template handles the entire OAuth flow automatically — you never have to write this code yourself.

The 3 Layers of Your App

Layer 1 — Your Server

Runs on Railway or Fly.io. This is your Node.js/Remix application. It holds all your business logic, your database, and your UI pages. You build and deploy this yourself.

Layer 2 — The Shopify Connection

Your app talks to Shopify using the access token. With this token, you can read products, create discount codes, fetch orders — anything the merchant gave you permission to do via the Admin API.

Layer 3 — Shopify Functions

Discount logic must run at checkout — in milliseconds, on Shopify's own servers. Shopify Functions are small programs you write and deploy to Shopify's infrastructure. They run there, not on your server. This is the most important concept for your discount app.

Your Tech Stack Explained

Every tool in your stack has a specific job. Understanding what each one is — and what it is NOT — will save you hours of confusion.

The House Building Analogy

House analogy	Your app
Blueprint + structure — decides how rooms connect	Remix — decides how pages connect, how data flows
Bricks — the basic building material	React — builds your UI components
Furniture — what the rooms look like inside	Polaris — pre-built UI components styled like Shopify
Filing cabinet — where records are physically kept	SQLite / PostgreSQL — where data is actually stored
Filing clerk — manages the cabinet for you	Prisma — talks to the database using JavaScript

Framework vs Library — The Key Difference

Library

A small, specific tool. You call it when you need it. It does one job. You are in full control of when and how to use it. You pick it up, use it, put it down.

Examples: [React](#), [Prisma](#), [Lodash](#), [Axios](#)

Framework

The entire structure your app lives inside. It decides how your app is organized, how pages work, how data moves around. You don't call a framework — you build inside it.

Examples: [Remix](#), [Next.js](#), [Laravel](#), [Django](#)

Remix

Framework

Remix is the full framework your app lives inside. It handles routing between pages, loading data from the server, handling form submissions, and managing sessions. Remix is built on top of React — React builds the visual components, Remix adds the server-side logic that React alone cannot do. Shopify officially recommends Remix for building apps.

React

UI Library

React is a JavaScript library for building user interfaces. You use it to create components — buttons, forms, cards, modals. It doesn't know anything about servers, databases, or routing. Remix handles all of that on top of React.

Polaris

UI Component Library

Polaris is Shopify's official component library. It gives you pre-built React components — cards, buttons, data tables, form inputs — that look exactly like Shopify's native admin design. Using Polaris makes your app feel native to Shopify, which merchants expect and Shopify reviewers look for.

Prisma

ORM / Database Toolkit

Prisma is NOT a database. It is a tool that sits between your JavaScript code and your actual database. Instead of writing raw SQL to query the database, you write normal JavaScript and Prisma translates it. Think of it as a remote control — the TV (database) stores everything, the remote (Prisma) is how you operate it from your code. Prisma also includes Prisma Studio — a visual browser-based view of all your tables.

SQLite

Database (Development)

SQLite is the actual database you use while building locally on your computer. It is just a single file stored on disk — zero setup, zero configuration. The Shopify CLI template configures this automatically. You use SQLite only during development.

PostgreSQL

Database (Production)

PostgreSQL is the proper database you switch to when you deploy to Railway. It lives in the cloud, handles thousands of concurrent connections, and is production-grade. Railway makes adding PostgreSQL to your app a one-click operation. You switch from SQLite to PostgreSQL by changing one line in your Prisma config.

The One-Sentence Summary

Remix is the house. **React** is the bricks. **Polaris** is the furniture. **Prisma** is the filing clerk. **SQLite/PostgreSQL** is the actual filing cabinet.

Hosting & Infrastructure

Shopify does NOT host your app. You need to host it yourself. Here is everything you need to know about your options.

Remember: Railway, Fly.io, Heroku, AWS, and GCP are all the same category of thing — they are all places where your app server lives. The difference is how easy they are to use and how much they cost.

Railway

[Start here](#)

Difficulty **Very Easy** **Cost** ~\$5/month **Setup time** 10 minutes

Connect your GitHub repo, click deploy. It handles everything else automatically. This is the most popular choice for Shopify apps today. [Start here](#).

Fly.io

[Also great](#)

Difficulty **Easy–Medium** **Cost** **Free tier available** **Setup time** 20 minutes

Similar to Railway but slightly more command-line based. A bit more control over your infrastructure. Very popular in the Shopify community.

Heroku

[Legacy choice](#)

Difficulty **Easy** **Cost** ~\$7/month min **Setup time** 15 minutes

Used to be the standard for beginners but removed their free tier in 2022. Costs more than Railway for the same capabilities. Still works fine.

AWS / GCP / Azure

[For later — when you're big](#)

Difficulty **Very Hard** **Cost** **Varies** **Setup time** **Days**

Extremely powerful but complex. You configure servers, load balancers, and networking yourself. Don't start here. Switch to these when you have thousands of paying merchants and need enterprise-grade infrastructure.

Important: Shopify ONLY hosts merchant stores. It does NOT host your app. Your app server must live on Railway (or similar) and Shopify loads it inside the merchant's admin via an iframe.

Your Learning Roadmap — 20 Steps

We go one step at a time. Each concept is explained first, then you write the code, then you confirm it works before moving to the next step.

Before You Start — What to Install

What to install	Why you need it
Node.js v18+	The runtime that executes your JavaScript server code
Shopify CLI	Official tool to scaffold and run Shopify apps locally
Shopify Partner account	Free at partners.shopify.com — needed to create apps
Development store	A free test Shopify store to install and test your app on
VS Code	The best code editor for this tech stack
Git + GitHub	Version control — start using this from day one

PHASE 0

Understand the Landscape

Week 1 · No coding yet

Step 1

How Shopify apps actually work

What is an embedded app? How does OAuth work? What is the App Store?

[Concept](#)

Step 2

The 3 Shopify APIs you will use

Admin API, Storefront API, and Shopify Functions — when to use each

[Concept](#)

Step 3

What is Shopify Functions?

The engine behind all custom discounts — the most critical concept in this app

[Concept](#)

Step 4

What is Remix?

Shopify's official framework — how it differs from plain React or Next.js

[Concept](#)**PHASE 1**

Set Up Your Development Environment

Week 1–2 · Setup

Step 5 Create Partner account and dev store partners.shopify.com — your free sandbox for testing everything you build Setup	Step 6 Install Node.js and Shopify CLI The two tools you must have before anything else can happen Setup
Step 7 Scaffold your first Shopify app Run `shopify app init` and understand every file and folder it creates Setup	Step 8 Understand the folder structure What is /app, /extensions, shopify.app.toml and why each one exists Concept

PHASE 2	Build Your First Working Screen	Week 2–3 · First real code
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Step 9 Remix routing basics How pages work in Remix — loaders, actions, and routes explained simply Code	Step 10 Shopify Polaris UI components The official component library — buttons, cards, forms, tables Code
Step 11 Your first Admin API call Fetch real data from the store — products, customers — using GraphQL Code	Step 12 Save data with Prisma + SQLite Store discount rules in a real database — start with SQLite locally Code

PHASE 3	Build Your First Discount Type	Week 3–5 · Core feature
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Step 13 Create your first Shopify Function The `shopify app generate extension` command — scaffold a discount function Build	Step 14 Write discount logic in a Function Input = cart data → Your logic → Output = discount targets. Build quantity breaks Build
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Step 15 Connect the UI to the Function Merchant fills a form → rules saved to DB → Function reads them at checkout Build	Step 16 Test it in your dev store Add products to cart, go to checkout, and see your discount apply in real time Build
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PHASE 4	Deploy, Publish, and Grow	Week 6+ · Go live
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Step 17 Deploy to production Host your app on Railway — get a real URL, connect it to Shopify Setup	Step 18 Add the billing system Shopify Billing API — charge merchants monthly, built-in, no Stripe needed Build
Step 19 Submit to the App Store What Shopify's review team checks and how to pass the review first time Setup	Step 20 Get your first installs and reviews Early growth tactics — Facebook groups, Reddit, cold outreach to merchants Growth

The App You Are Building — DiscountLab

The complete discount engine for Shopify. Every promotion type in one app, built for merchants who have outgrown Shopify's native discounts.

Core Positioning

One app. Every promotion type. Built for merchants who outgrow Shopify's 4 native discount types.

All Discount Modules

Volume & Tiered

- Quantity breaks — buy 3 get 10%, buy 5 get 20%
- Cart value tiers — spend \$50 get 5%, spend \$100 get 15%
- Subscribe & save — % off on recurring orders

Bundling

- Fixed-price bundles — any 3 items for \$29
- Build-your-own bundle — customer picks N items
- Mix & match across collections

Flash & Time-Based

- Countdown timer discounts — offer expires in X hours
- Happy hour pricing — active only between set hours
- Early bird / limited-time launch pricing

Gift With Purchase

- Spend \$X, get a free gift auto-added to cart
- Gift choice modal — customer picks from 2–3 options
- Threshold-based free shipping unlock

Post-Purchase

- Thank-you page one-click upsell with discount
- Reorder discount — email triggered X days later
- Cross-sell bundles after purchase

Customer Segments

- VIP / tag-based pricing — Gold, Staff, Wholesale
- First-time buyer welcome discount
- Win-back discount for inactive 60/90/120 day customers
- Birthday discount — auto-sends in birthday month

Referral & Social

- Refer-a-friend — give \$10, get \$10
- Influencer codes with usage caps and expiry
- Share-to-unlock discounts

Flat-Price Store ■

- Everything at \$99 — regardless of original price
- Perfect for clearance events and flash sales
- One of the most requested discount types — underserved

Wholesale / B2B

- Company-specific price lists
- Customer tag-based pricing — no code needed
- Net terms discounts (Net-30 = 2% off)

Cart Behavior

- Exit-intent popup discount
- Abandoned cart recovery discount
- Minimum quantity per product discounts

Freemium Pricing Model

Free \$0 / month	Growth \$19 / month	Pro \$49 / month
· 1 active discount rule · Quantity breaks (basic) · Cart value threshold · Flat-price store (limited) · No analytics	· Up to 10 active rules · All volume & tiered types · Bundling (fixed + BYOB) · Flash / countdown timers · Gift with purchase · Basic analytics dashboard	· Unlimited active rules · All discount modules · Customer segmentation · Referral & influencer codes · Post-purchase offers · Wholesale / B2B pricing · Advanced analytics + CSV export

The Most Important Technical Constraint

One Shopify Function per checkout

Shopify only allows one discount function of each type to run at checkout. This means your app must combine ALL of a merchant's active discount rules into a single Shopify Function that evaluates them all together. This central rules engine is the core architectural challenge of the app — and what separates well-built discount apps from buggy ones.

This guide will be updated at the end of every learning session. By the time we reach Step 20, this document will be a complete reference for everything you built and learned.